

ISLAMIC UNIVERSITY OF TECHNOLOGY

PROJECT REPORT: PHASE C,D & E (SW)

Team Name: ReVOLT

Team Members:

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Phase C: Control Units

1. Instruction Register
2. Controller Sequencer

Phase D:

1. Arithmetic and Logic Unit (ALU)
2. Integration with 8-bit Bus
3. Output Unit

Phase E:

1. Complete Integration of SAP-1

Phase C

1. Instruction Register

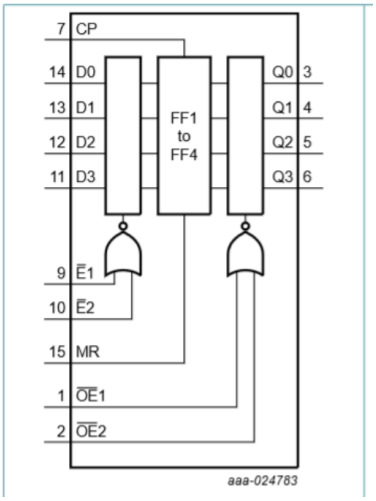
The instruction register is used for receiving and storing six bits of instructions stored on the bus from RAM. Two segments of approximately 3,3 bits are then drawn from the contents of the instruction register. The upper three bits of the 'upper 3' string will go to the controller sequencer while the Lower Three Bits should be passed on to the bus. When the computer is launched, a reset pin will be needed to reset your instruction register. Description of the control bits –

Pin	Name	Function
II	Instruction Register In	Writes the current values of the bus into the instruction register.
IO	Instruction Register Out	Sends the stored data (only lower 3 bits) onto the bus. Output pins are disconnected when IO is low.

Description of used ICs:

a) IC 74HC173 – Quad D-type flip-flop positive-edge trigger:

The device features clock (CP), master reset (MR), two input enable (E1, E2) and two output enable (OE1', OE2') inputs. When the input enables are LOW, the outputs Qn will assume the state of their corresponding Dn inputs that meet the set-up and hold time requirements on the LOW-to-HIGH clock (CP) transition. A HIGH on either input enable will cause the device to go into a hold mode, outputs halt their previous state independently of clock and data inputs. A HIGH on MR forces the outputs LOW independently of clock and data inputs. Operation of the output enable inputs does not affect the state of the flip-flops.



b) IC 74HC245 – Octal 3−state noninverting bus transceiver:

It is a 20-pin IC, the detail of all pins is mentioned below.

Pin#01 DIRECTION CONTROL PIN: The direction control pin (DIR) is used to decide the direction of communication whether data will from transport A to B or from B to A.

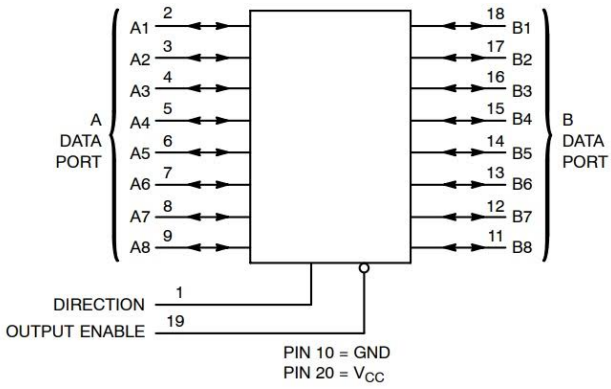
Pin#02 to 09 A0-A7: These are input pins (as used here) that also behaves as output depending on the direction control input.

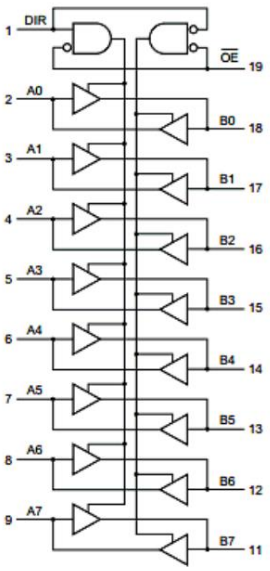
Pin#10 Ground

Pin#11 to 18 B0-B7: These are output pins (as used here) that also behaves as input depending on the direction control input.

Pin#19: It is the active-low output enable pin. It controls the output by enabling or disabling the inputs. This pin is used for cascading.

In the pin 1, always 1 is given to denote that data is going from A data port to B data port. The CO and other inputs and outputs are connected to the pins according to the diagrams –





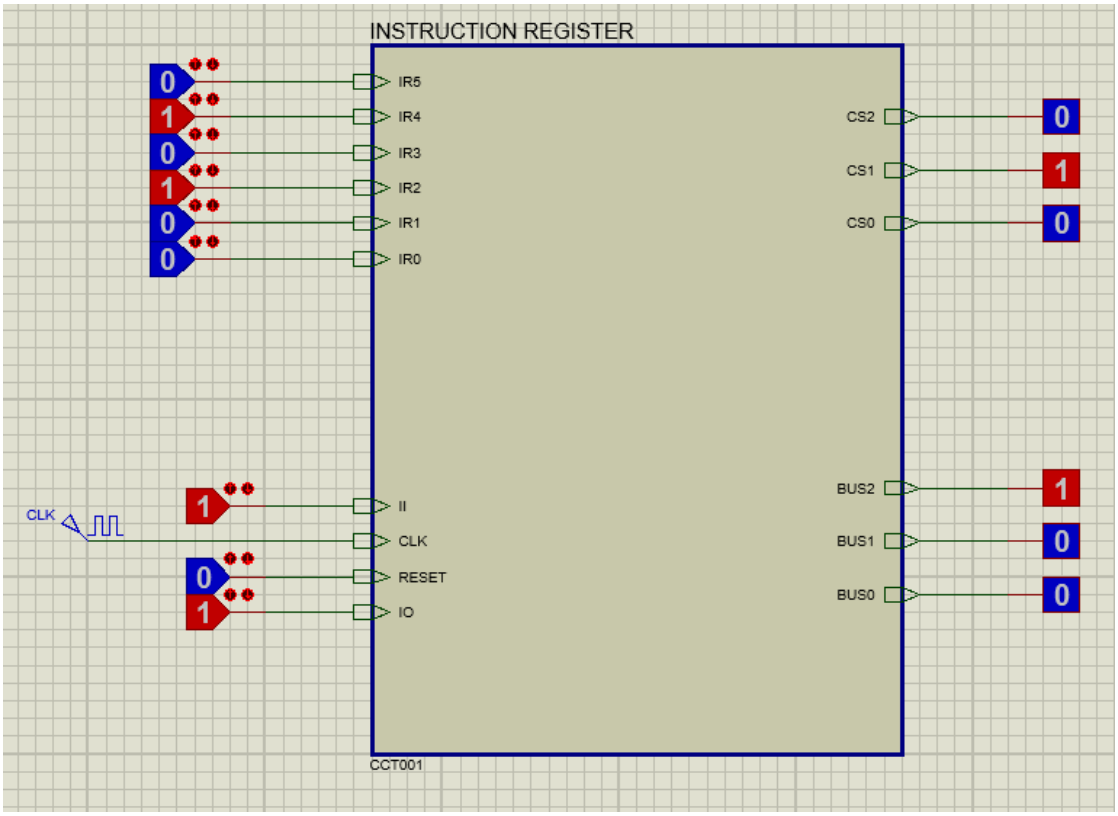
Design:

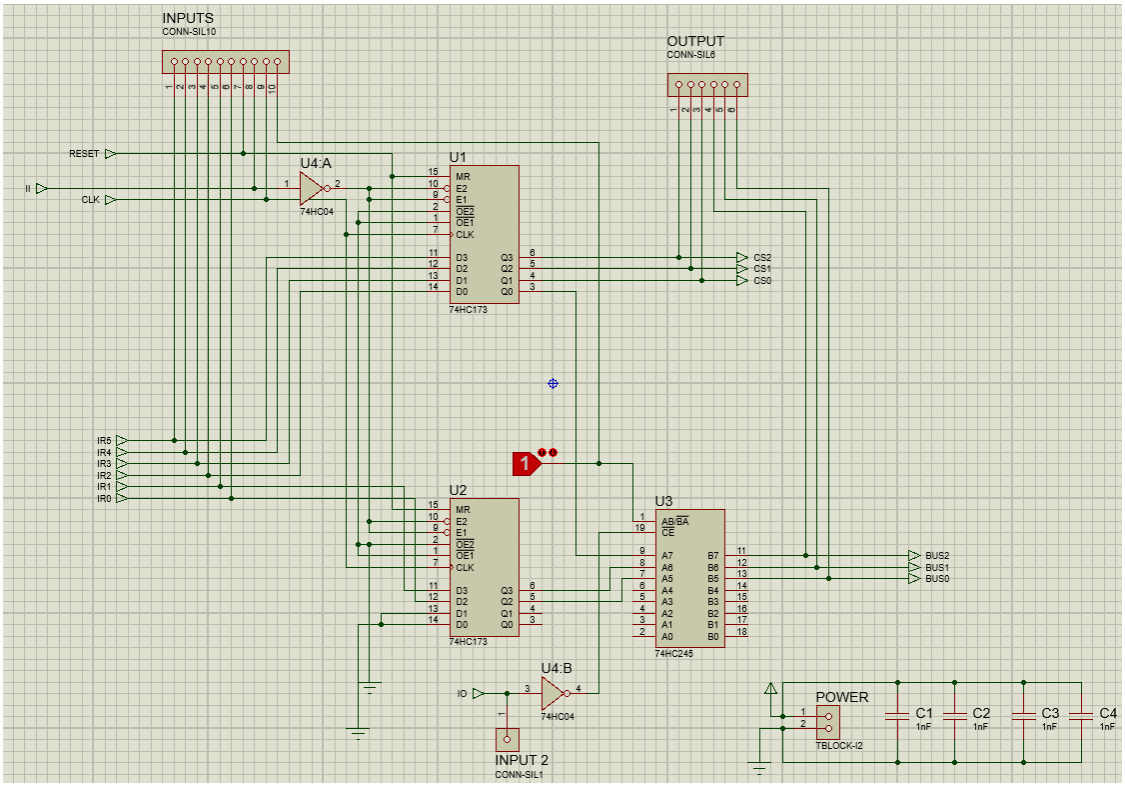
There are two 74HC173 ICs which works as 4-bit D registers. Each contains 4 input data pins as well as 4 output pins. 4 inputs are given and 4 outputs are gotten from the first IC. And 2 inputs are given and 2 outputs are gotten from the second IC. Thus 6 inputs are given and 6 outputs are gotten from the 74HC173 ICs.

The 3 upper bits outputs (CS0, CS1, CS2) goes to controller sequencer directly. In this case, we will get the output we give in the input. And the 3 lower bits outputs (BUS1, BUS2, BUS3) are passed through 74HC245, bus transceiver. In this case, the output enable pin (19) is connected AO control bit. It is an active high pin, as we used a NOT gate in the 19th pin. When IO=0, output passes and when IO=1, we don't get any output in bus.

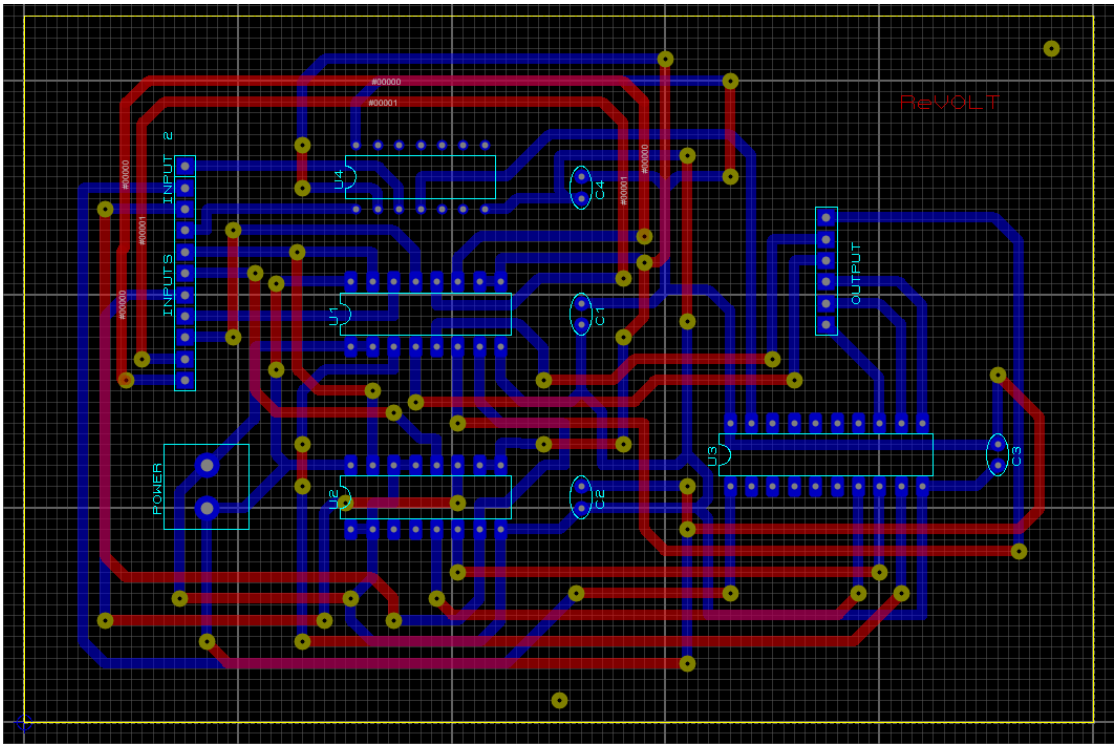
The 4 other input pins are II, CLK, RESET, IO. II is active high, as we used a NOT gate in the 9 & 10th pin. When II=1, we get the current values or the changes in outputs with the change in input, when II=0, we don't get the current values in outputs. RESET is active high. When RESET=1, the outputs reset to 000. IO is the output pins that controls bus outputs. It is active low. When IO=1, we get outputs in the bus, when IO=0, we don't get outputs in the bus.

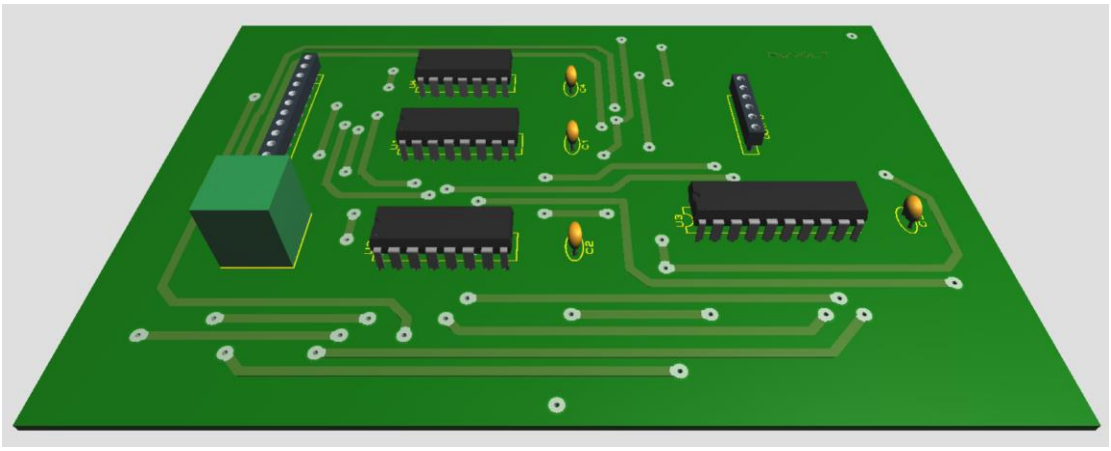
Schematic Layout:

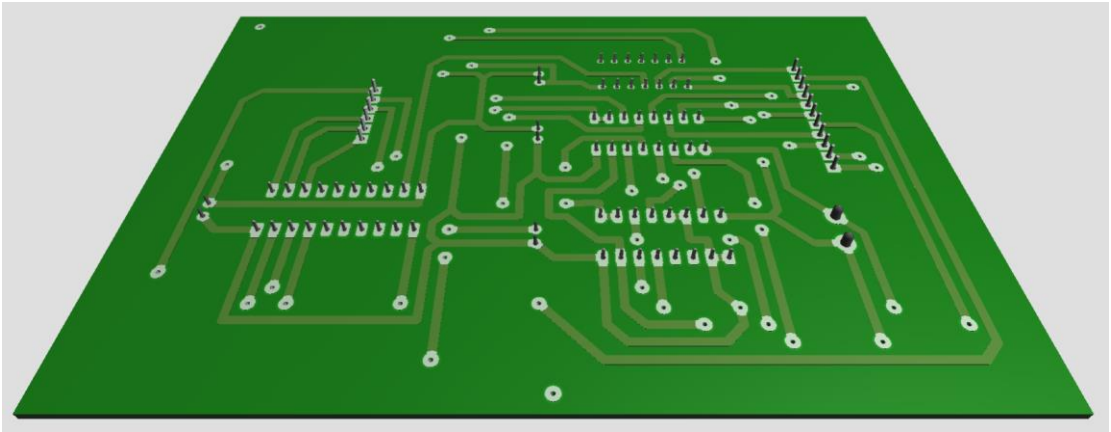




PCB Layout:







2. Controller-Sequencer:

Controller-sequencer generates necessary control signals for each block so that actions occur in a desired sequence. The controller-sequencer receives the instruction from the Instruction register and sends out signals to execute our desired process that can control our SAP and makes sure things happen only when they are supposed to. 12 control bits come out of the controller-sequencer block. These control bits determine how all other blocks will react to the next positive CLK edge. So, controller sequencer is designed in such a way so that correct control bits are already available to all the blocks before the next positive CLK edge.

12 control bits are –

CON = CE CO MI RO II IO AI AO SU EO BI OI

Besides these 12 signals, the controller sequencer also generates a HLT (halt) signal, which can halt the process of computer by stopping the main clock.

In each of our instruction cycles, there are 5 T-states. We reduced one T-state. We combined the T2 and T3 states mentioned in the project manual. T-state means the time required to complete each microinstruction. For each instruction, different control bits are active in different t-state. We have implemented a ring counter to generate our T-states. And we used basic logic gates or the selection of our control bits during different T-states.

T state	LDA(000)	ADD(001)	SUB(010)	OUT(011)
T1	CO,MI	CO,MI	CO,MI	CO,MI
T2	CE,RO,II	CE,RO,II	CE,RO,II	CE,RO,II
T3	IO,MI	IO,MI	IO,MI	AO,OI
T4	RO,AI	RO,BI	RO,BI	-
T5	-	EO,AI	EO,AI,SU	-

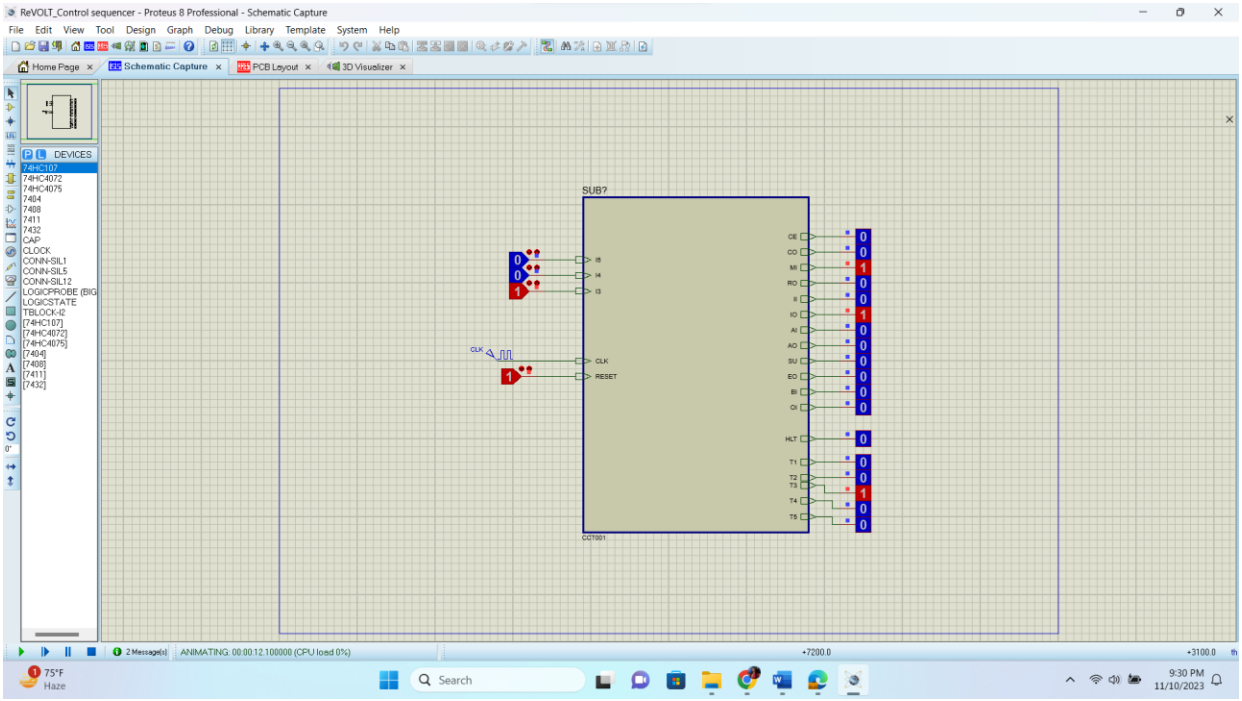
The control bit equations are:

[Here, +=AND , .=OR]
CE= T2
CO= T1
MI= T1+ T3.LDA + T3.ADD+ T3.SUB
RO= T2+ T4.LDA+ T4.ADD +T4.SUB
II= T2
IO= T3.LDA+ T3.ADD+ T3.SUB
AI= T4.LDA+ T5.ADD+ T5.SUB
AO= T3.OUT
SU= T5.SUB
EO= T5.ADD+ T5.SUB
BI= T4.ADD+ T4.SUB
OI= T3.OUT

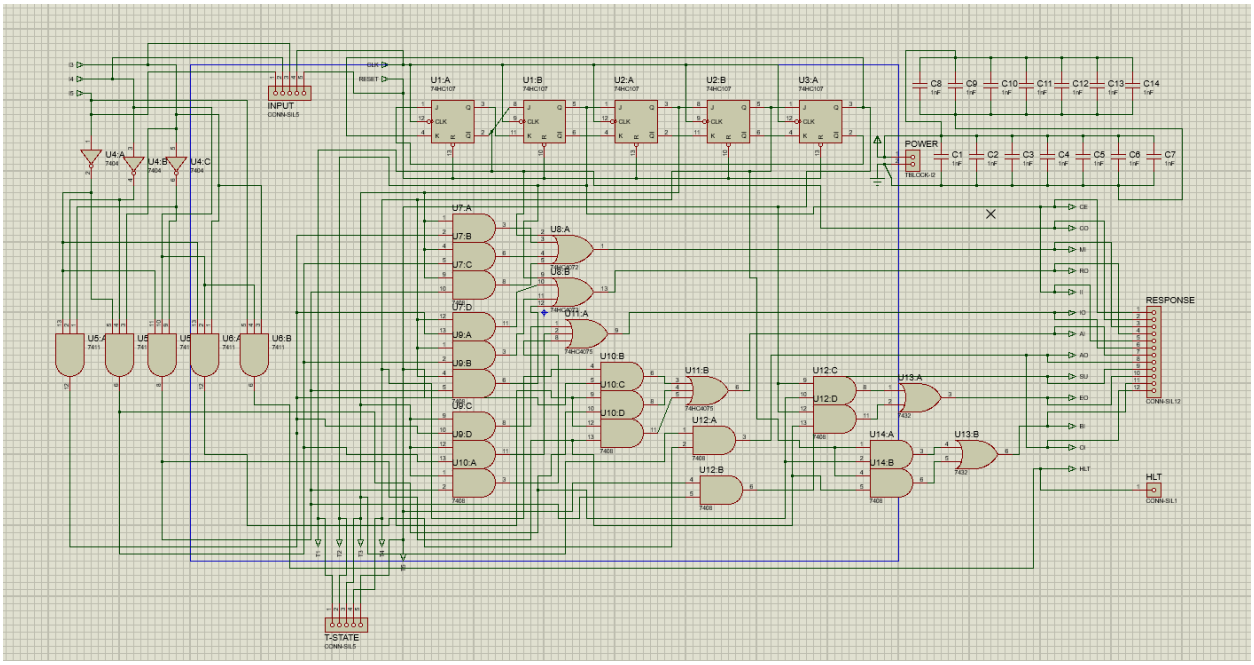
HLT= HLT

Simulation:

Parent sheet

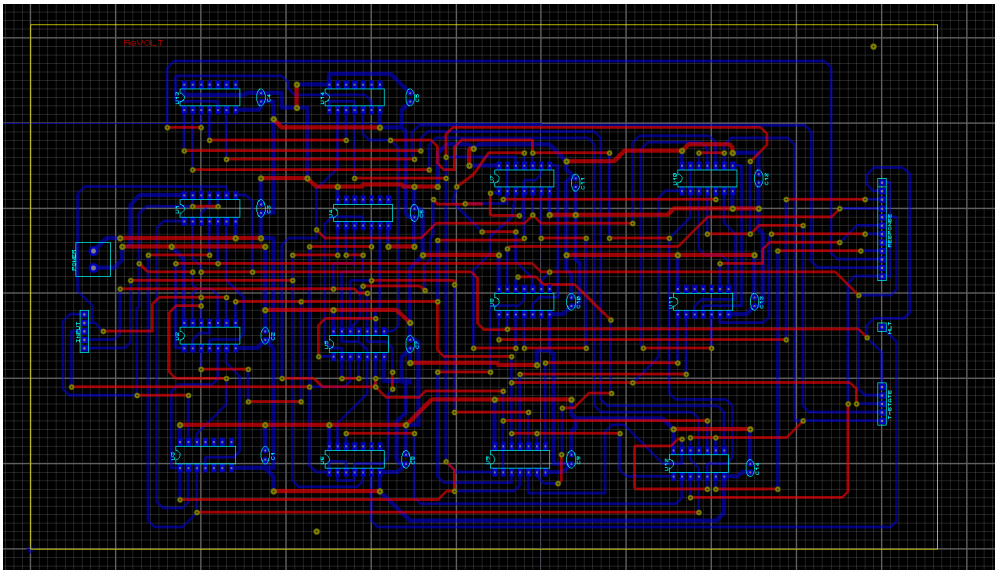


Inside child sheet

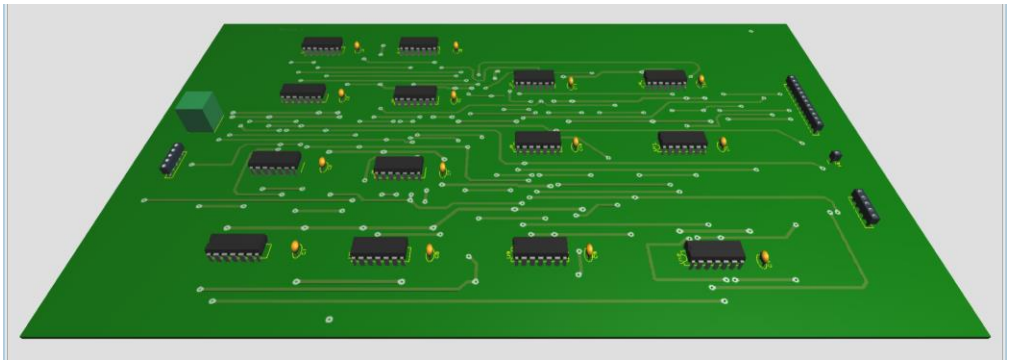


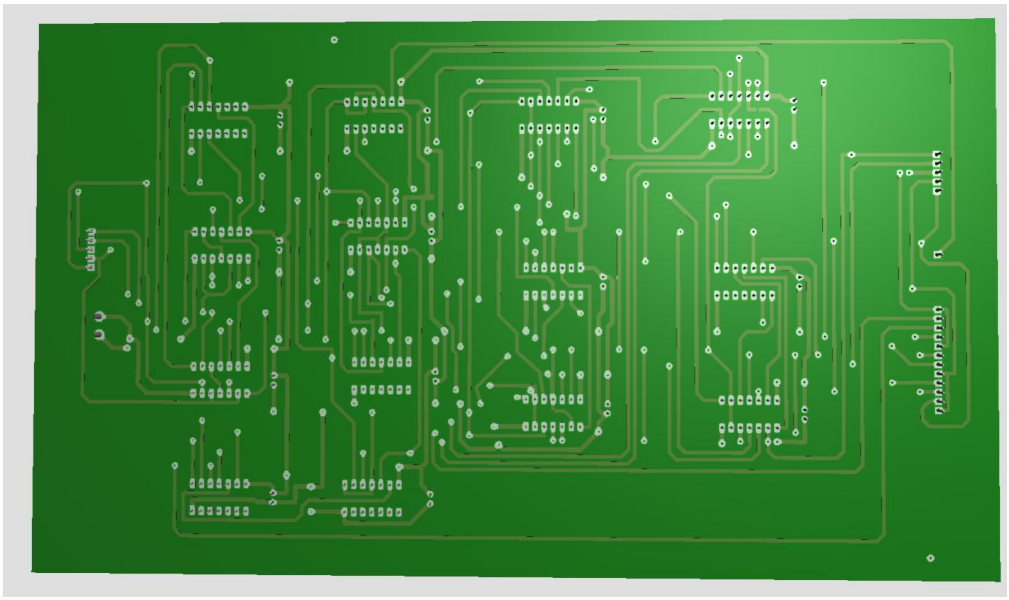
PCB Layout

Design



In 3D visualizer,





Phase D

1. Arithmetic and Logic Unit (ALU)

ALU is capable of 6-bit addition and subtraction. For addition and subtraction, 2's complement approach is used. Control bits of ALU –

Pin	Name	Function
SU	Addition/Subtraction Select	When SU=0, addition operation and when SU=1, subtraction operation will be executed.
EO	ALU Out	Puts the ALU output onto the bus. Output of ALU should be disconnected when EO is low.

Description of used ICs:

a) IC 74HC283 – 4-bit binary full adder:

The 74HC283 adds two 4-bit binary words (An plus Bn) plus the incoming carry (CIN). The binary sum appears on the sum outputs (S1 to S4) and the out-going carry (COUT) according to the equation:

CIN + (A1 + B1) + 2(A2 + B2) + 4(A3 + B3) + 8(A4 + B4) = S1 + 2S2 + 4S3 + 8S4 + 16COUT

Pin description

Symbol	Pin	Description
S2	1	sum output 2
B2	2	B operand input 2
A2	3	A operand input 2
S1	4	sum output 1
A1	5	A operand input 1
B1	6	B operand input 1
CIN	7	carry input
GND	8	ground (0 V)
COUT	9	carry output
S4	10	sum output 4
B4	11	B operand input 4
A4	12	A operand input 4
S3	13	sum output 3
B3	14	A operand input 3
A3	15	B operand input 3
V_{CC}	16	positive supply voltage

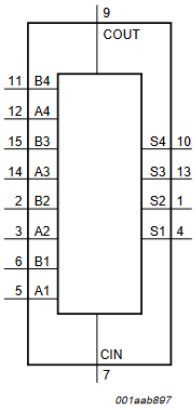


Fig 1. Functional diagram

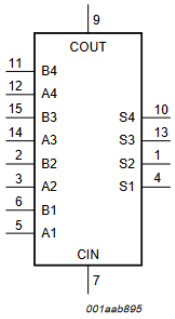


Fig 2. Logic symbol

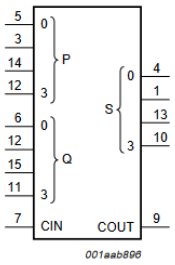


Fig 3. IEC logic symbol

b) IC 74HC245 – Octal 3−state noninverting bus transceiver:

Described earlier.

Design:

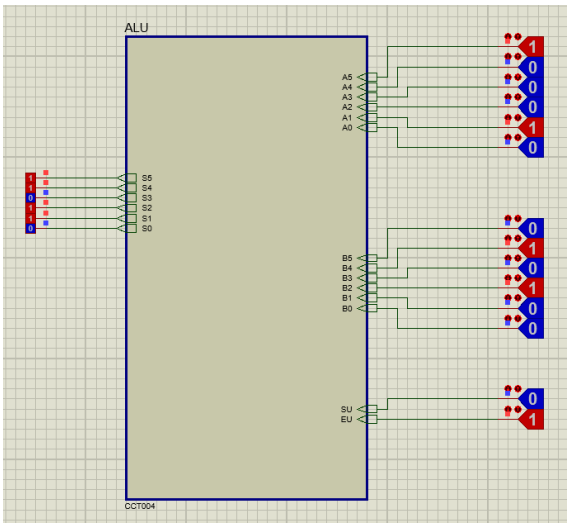
2 adder IC was used. In the first adder IC, 4 & 4 total 8 inputs (A0, A1, A2, A3, B0, B1, B2, B3) were connected. And in the second adder IC, 2 & 2 total 4 inputs (A4, A5, B4, B5) were connected. And S0, S1, S2, S3 outputs were gotten from the first adder and S4, S5 outputs were gotten from the second adder. The outputs were passed through bus transceiver IC to connect EO pin to control output passing.

The B0, B1, B2, B3, B4, B5 inputs were each connected with SU pin through an X-OR gate. This was to determine the addition or subtraction operation. When SU=0, according to the characteristics of X-OR gate, the inputs remain as it is. But when SU=1, according to the characteristics of X-OR gate, the inputs are complemented.

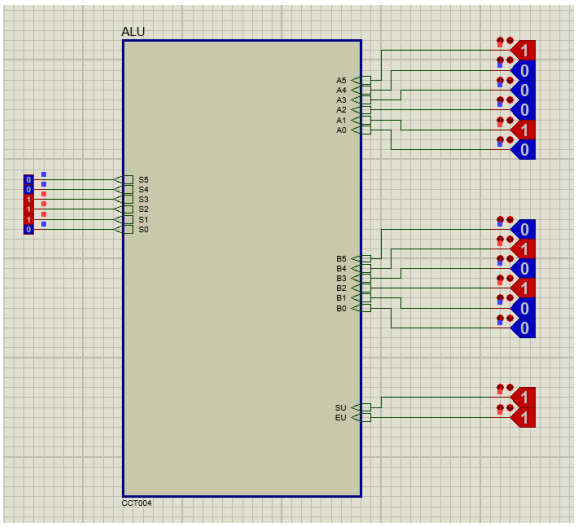
SU pin is also connected to 7th pin (C_{in}) of the first adder IC. So that when SU=1, in case of subtraction, 1 is added with LSB for 2's complement. The 9th pin (C_{out}) of the first Adder IC is connected to the 7th pin (C_{in}) of the second Adder IC.

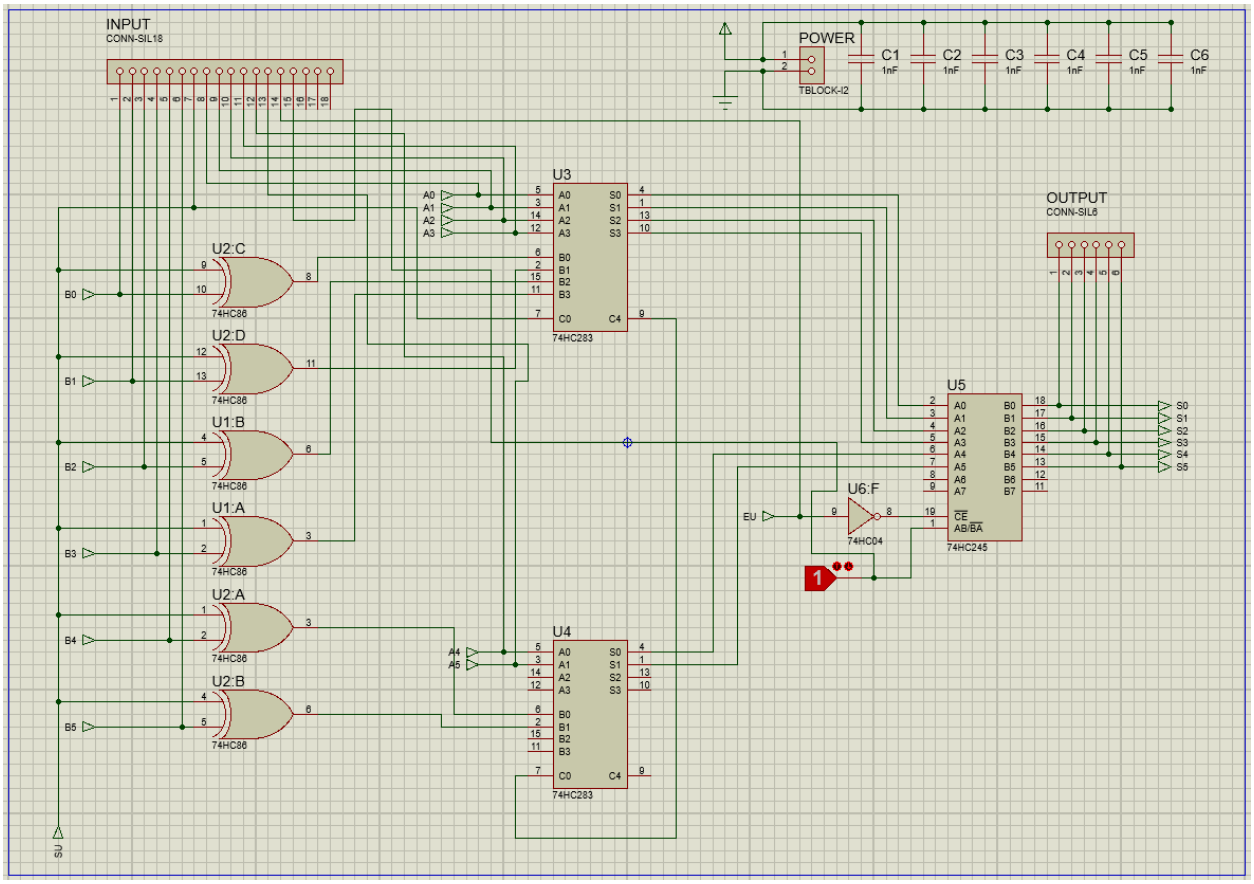
Schematic Layout:

Addition (SU=0)

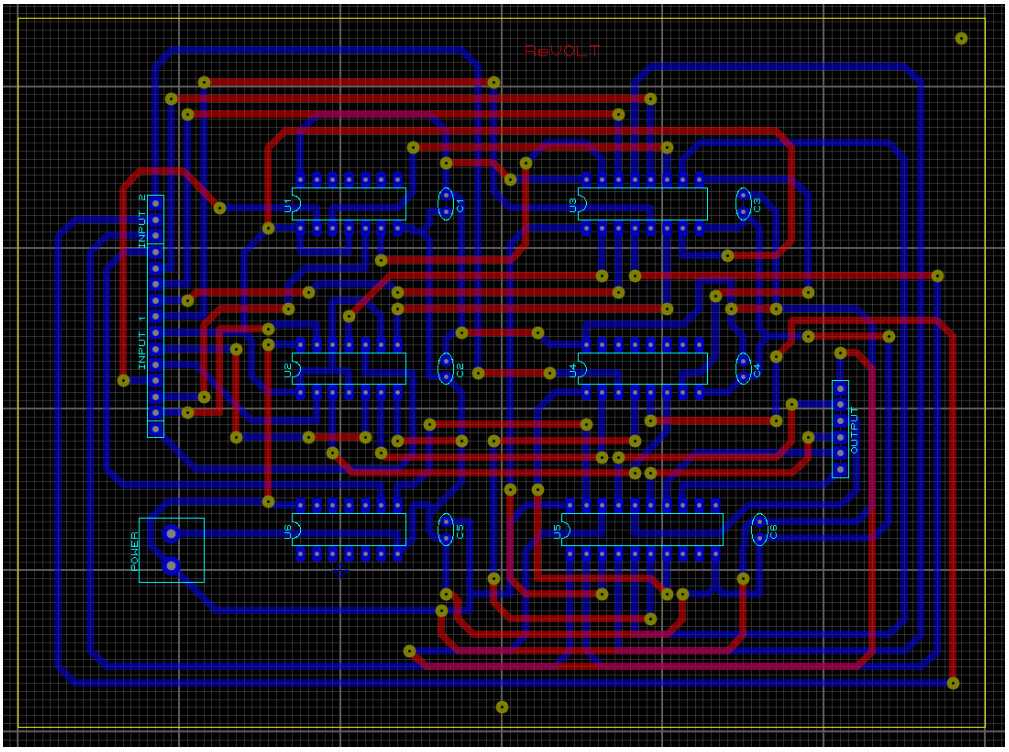


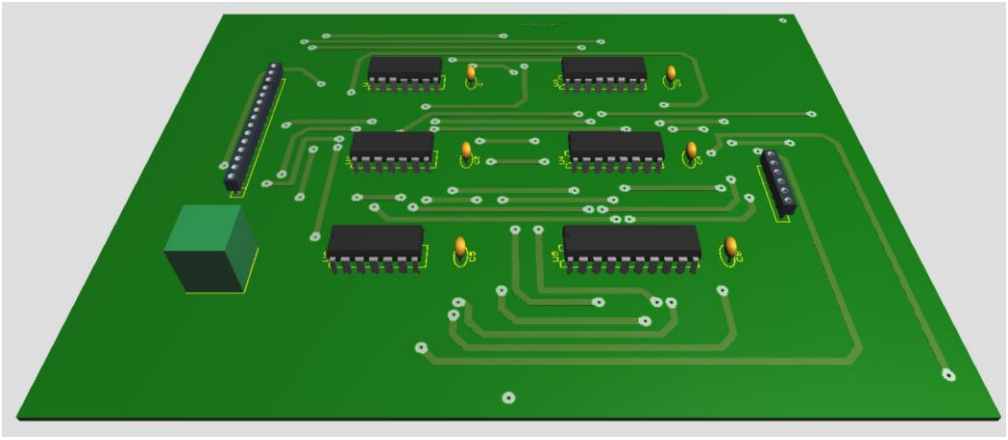
Subtraction (SU=1)

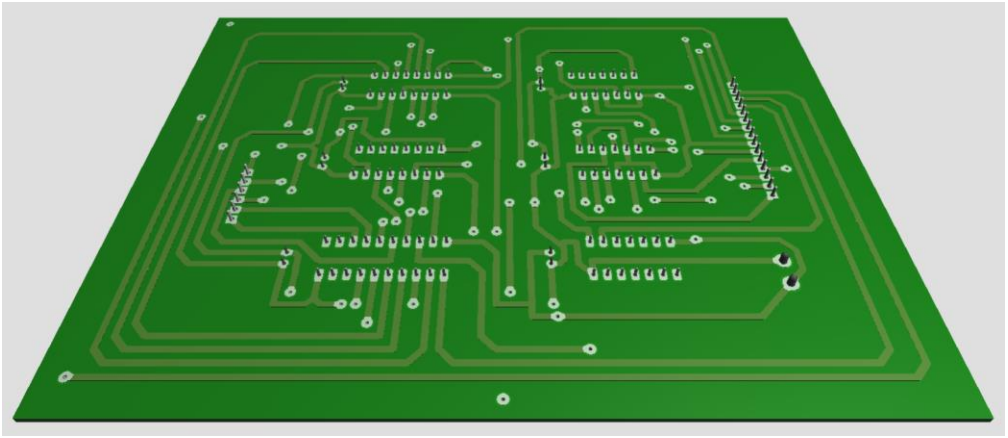




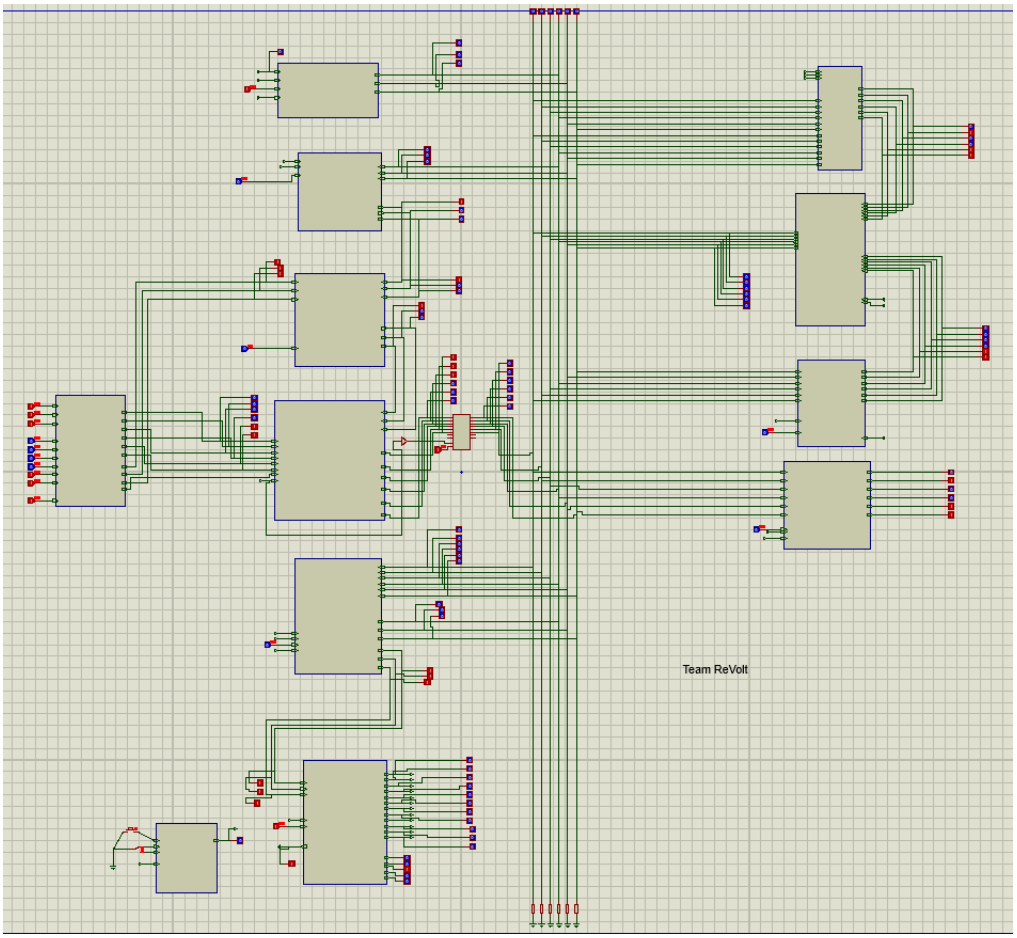
PCB Layout:


FeuVOLT
INPUT 2
INPUT 1
POWER
OUTPUT
U1
U2
U3
U4
U5
U6
C1
C2
C3
C4
C5
C6

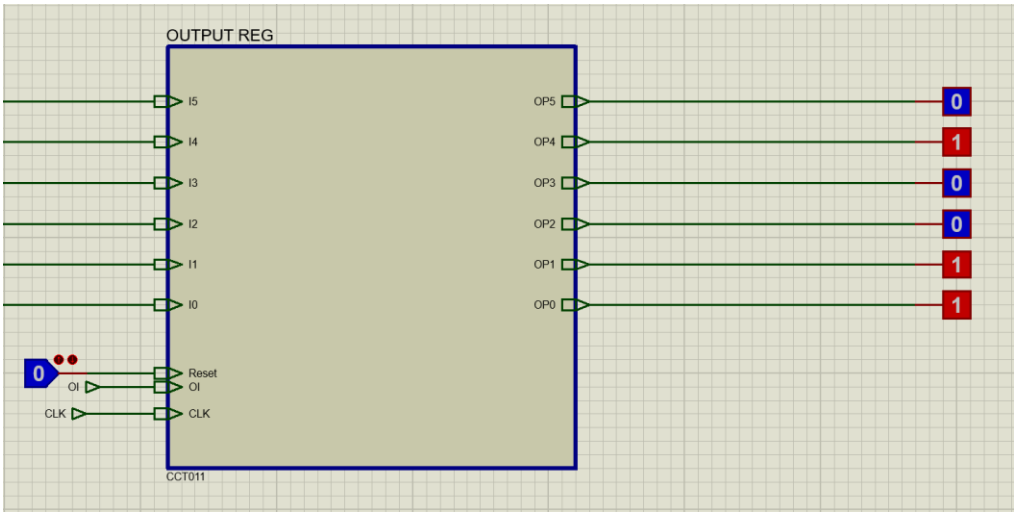




2. Integration with 8-bit Bus

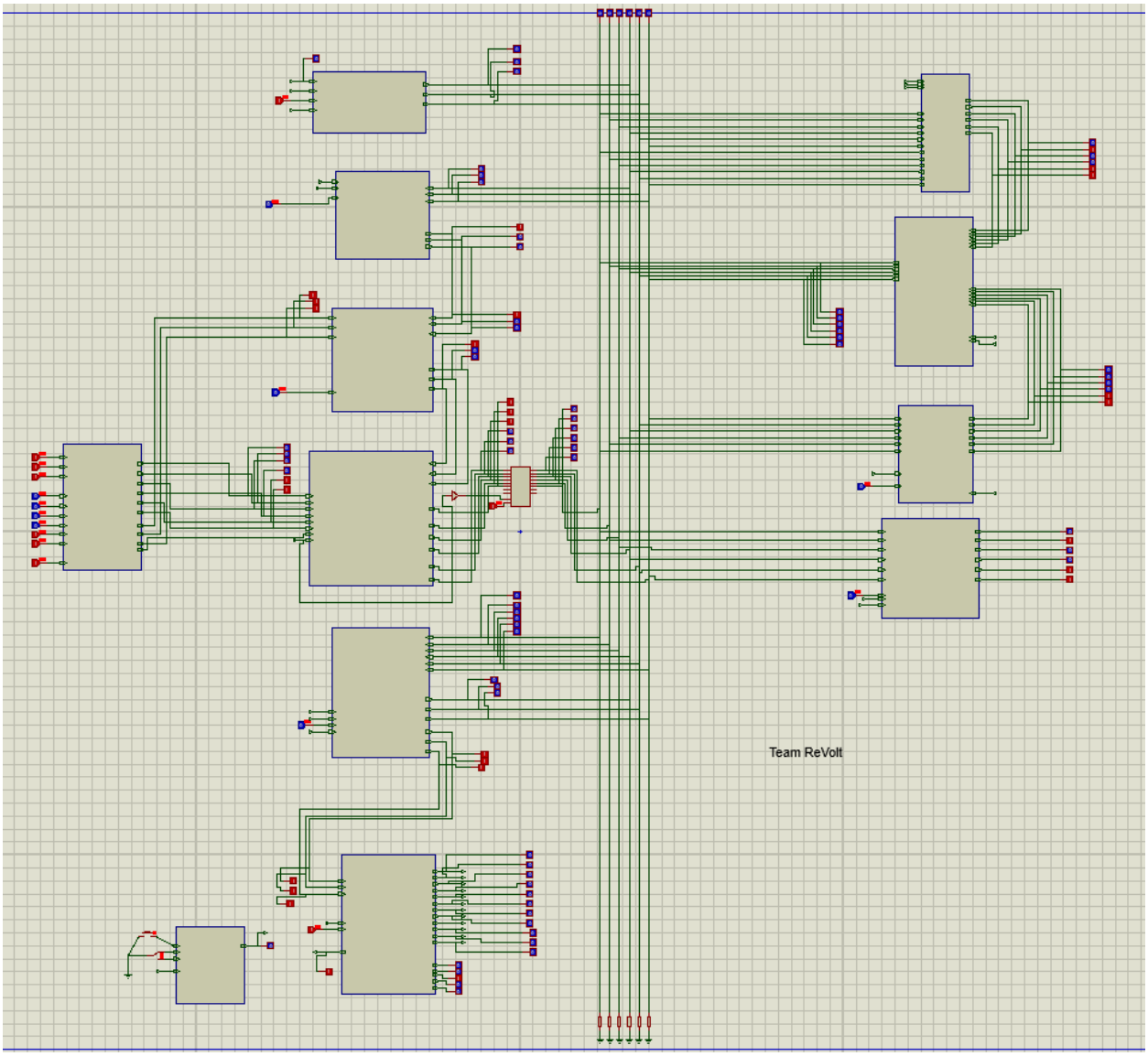


3. Output Unit



Phase E

Complete Integration of SAP-1



Contribution:

210021118: Design of ALU; Report Making

210021139: Instruction Register, PCB of Instruction Register & ALU; Report Making

210021212: Integration

210021214: Integration

210021238: Design of Instruction Register, ALU; Report Making

210021242: Design & PCB of Controller Sequencer

210021256: Cross-check